Decent Coaching Classes

IX CBSE

Mind Your Math VIII — Answer Key & Marking Scheme

Time : 3 Hours **Maximum Marks : 80**

SECTION A — Multiple Choice Questions 20 × 1 = 20

1. The point (−3, 4) lies in the

   (a) I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant

   Ans: (b) II quadrant [x < 0 and y > 0, so it lies in the II quadrant.]

2. The distance of the point (3, −4) from the x-axis is

   (a) 4 units (b) 3 units (c) 5 units (d) −4 units

   Ans: (a) 4 units [Distance from the x-axis = |y| = 4.]

3. Which of the following points lies on the y-axis?

   (a) (4, 0) (b) (2, 2) (c) (0, −3) (d) (−1, 5)

   Ans: (c) (0, −3) [Every point on the y-axis has x = 0.]

4. The zero of the linear polynomial p(x) = 2x − 6 is

   (a) −3 (b) 6 (c) 12 (d) 3

   Ans: (d) 3 [2x − 6 = 0 ⇒ x = 3.]

5. Which of the following is a linear polynomial in x?

   (a) 3x − 5 (b) $x^2 + 1$ (c) 1/x + 2 (d) √x

   Ans: (a) 3x − 5 [Only 3x − 5 has highest power 1 with whole-number powers.]

6. If x = 2 is a zero of the polynomial 3x − k, then k =

   (a) 2 (b) 3 (c) 6 (d) −6

   Ans: (c) 6 [3(2) − k = 0 ⇒ k = 6.]

7. Which of the following is an irrational number?

   (a) √16 (b) √7 (c) 0.25 (d) 22/7

   Ans: (b) √7 [√7 is non-terminating and non-repeating; √16 = 4.]

8. $2^3 \times 2^4$ is equal to

   (a) 2^{12} (b) 4^7 (c) 2^7 (d) 2

   Ans: (c) 2^7 [$a^m \times a^n = a^{m+n} \Rightarrow 2^7$.]

9. The value of $81^{1/4}$ is

   (a) 9 (b) 3 (c) 27 (d) 1/3

   Ans: (b) 3 [$81 = 3^4$, so $81^{1/4} = 3$.]

10. On rationalising the denominator, 1/(√3 − 1) becomes

    (a) (√3 − 1)/2 (b) √3 + 1 (c) (√3 + 1)/4 (d) (√3 + 1)/2

    Ans: (d) (√3 + 1)/2 [Multiply by (√3 + 1): (√3 + 1)/(3 − 1) = (√3 + 1)/2.]

11. (x + 3)(x − 3) =

    (a) $x^2 - 9$ (b) $x^2 + 9$ (c) $x^2 - 6x + 9$ (d) $x^2 + 6x + 9$

    Ans: (a) $x^2 - 9$ [$(a + b)(a - b) = a^2 - b^2$.]

12. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 =$

(a) 37 (b) 19 (c) 25 (d) 49

Ans: (c) 25 [$x^2 + y^2 = (x + y)^2 - 2xy = 49 - 24 = 25$.]

13. Using a suitable identity, $101 \times 99 =$

(a) 10001 (b) 9999 (c) 9899 (d) 10099

Ans: (b) 9999 [$(100 + 1)(100 - 1) = 100^2 - 1 = 9999$.]

14. The angle subtended by a diameter of a circle at any point on the circle is

(a) 90° (b) 60° (c) 120° (d) 180°

Ans: (a) 90° [Angle in a semicircle is a right angle.]

15. The longest chord of a circle is its

(a) radius (b) arc (c) diameter (d) secant

Ans: (c) diameter [The diameter is the longest chord.]

16. An arc of a circle subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is

(a) 100° (b) 200° (c) 25° (d) 50°

Ans: (d) 50° [Angle at the circle = ½ × angle at the centre = 50°.]

17. The circumference of a circle of radius 7 cm is (take $\pi = 22/7$)

(a) 44 cm (b) 154 cm (c) 49 cm (d) 22 cm

Ans: (a) 44 cm [$2\pi r = 2 \times 22/7 \times 7 = 44$ cm.]

18. The perimeter of a rectangle of length 12 cm and breadth 8 cm is

(a) 96 cm (b) 20 cm (c) 80 cm (d) 40 cm

Ans: (d) 40 cm [$2(12 + 8) = 40$ cm.]

19. A die is rolled once. The probability of getting a prime number is

(a) 1/6 (b) 1/2 (c) 1/3 (d) 2/3

Ans: (b) 1/2 [Primes: 2, 3, 5 → 3/6 = 1/2.]

20. The 5th term of the AP 2, 5, 8, 11, … is

(a) 11 (b) 17 (c) 14 (d) 20

Ans: (c) 14 [$a = 2, d = 3$: $t_5 = 2 + 4 \times 3 = 14$.]

SECTION B — Very Short Answer Questions 5 × 2 = 10

21. Write the coordinates of the point which lies 3 units to the right of the y-axis and 5 units below the x-axis.

The point is $(3, -5)$.

22. If $p(x) = 2x - 3$, find $p(-1)$ and the zero of $p(x)$.

$p(-1) = 2(-1) - 3 = -5$ (1)

Zero: $2x - 3 = 0 \Rightarrow x = 3/2$ (1)

23. Simplify: $27^{2/3} \times 5^0$.

$27^{2/3} = (3^3)^{2/3} = 3^2 = 9$ (1)

$5^0 = 1$, so the value $= 9 \times 1 = 9$ (1)

24. A bag contains 3 red and 5 blue balls. One ball is drawn at random. Find the probability that it is red.

Total outcomes = 8, favourable = 3 (1)

P(red) = 3/8 (1)

25. Find the 10th term of the AP 7, 10, 13, …

$a = 7, d = 3$ (1)

$t_{10} = 7 + 9 \times 3 = 34$ (1)

SECTION C — Short Answer Questions 6 × 3 = 18

26. Express 0.777… in the form p/q, where p and q are integers and q ≠ 0.

Let $x = 0.777\ldots$ (i)

$10x = 7.777\ldots$ (ii) (1)

(ii) − (i): $9x = 7$ (1)

$x = 7/9$ (1)

27. Using a suitable identity, evaluate 102^2.

$102^2 = (100 + 2)^2$ (1)

$= 100^2 + 2(100)(2) + 2^2$ (1)

$= 10000 + 400 + 4 = 10404$ (1)

28. A chord of length 16 cm is drawn in a circle of radius 10 cm. Find the distance of the chord from the centre.

The perpendicular from the centre bisects the chord, so half-chord = 8 cm (1)

Radius^2 = half-chord^2 + distance^2: $10^2 = 8^2 + d^2$ (1)

$d^2 = 36 \Rightarrow d = 6$ cm (1)

29. A rectangular field is 40 m long and 25 m wide. A path 2 m wide runs all around it on the outside. Find the area of the path.

Area of field = $40 \times 25 = 1000$ m^2 (1)

Outer rectangle = $(40 + 4) \times (25 + 4) = 44 \times 29 = 1276$ m^2 (1)

Area of path = $1276 - 1000 = 276$ m^2 (1)

30. A die is thrown once. Find the probability of getting (i) an even number, (ii) a number greater than 4, (iii) a number less than 7.

Total outcomes = 6

(i) Even: 2, 4, 6 → P = 3/6 = 1/2 (1)

(ii) Greater than 4: 5, 6 → P = 2/6 = 1/3 (1)

(iii) All outcomes are less than 7 → P = 1 (1)

31. Which term of the AP 5, 11, 17, 23, … is 101?

$a = 5, d = 6, t_n = 101$ (1)

$5 + (n - 1) \times 6 = 101 \Rightarrow (n - 1) \times 6 = 96$ (1)

$n - 1 = 16 \Rightarrow n = 17$. So 101 is the 17th term. (1)

SECTION D — Long Answer Questions 4 × 5 = 20

32. Draw the graph of the linear equation $y = 2x - 1$.

(a) Complete the table of values for $x = -1, 0, 1, 2, 3$. [2]

(b) Plot the points on graph paper and join them to get the line. [2]

(c) From your graph, read the value of x when $y = 1$. [1]

(a) $y = 2x - 1$: $x = -1 \to y = -3$; $x = 0 \to y = -1$; $x = 1 \to y = 1$; $x = 2 \to y = 3$; $x = 3 \to y = 5$ (2)

(b) Points (−1, −3), (0, −1), (1, 1), (2, 3), (3, 5) plotted correctly (1); straight line through them (1)

(c) The line meets $y = 1$ at $x = 1$ (1)

33. If $x + 1/x = 4$, find the value of

(i) $x^2 + 1/x^2$ [3]

(ii) $x^4 + 1/x^4$ [2]

(i) $(x + 1/x)^2 = x^2 + 2 + 1/x^2$ (1) $\Rightarrow$ $16 = x^2 + 1/x^2 + 2$ (1) $\Rightarrow$ $x^2 + 1/x^2 = 14$ (1)

(ii) $(x^2 + 1/x^2)^2 = x^4 + 2 + 1/x^4$ (1) $\Rightarrow$ $196 = x^4 + 1/x^4 + 2 \Rightarrow x^4 + 1/x^4 = 194$ (1)

34. O is the centre of a circle and AB is a chord with $\angle AOB = 110°$. A point C lies on the major arc and a point D lies on the minor arc AB. Find (i) $\angle ACB$, (ii) $\angle ADB$, (iii) $\angle OAB$.

(i) Angle at the circle = ½ × angle at the centre: $\angle ACB = ½ \times 110° = 55°$ (2)

(ii) ACBD is a cyclic quadrilateral, so $\angle ADB = 180° - 55° = 125°$ (2)

(iii) $OA = OB$, so $\angle OAB = (180° - 110°)/2 = 35°$ (1)

35. A circular garden of radius 14 m is surrounded by a path 7 m wide. Find the area of the path and the cost of paving it at ₹ 50 per m^2. (Use $\pi = 22/7$)

Outer radius = 14 + 7 = 21 m (1)

Area of path = $\pi(21^2 - 14^2) = (22/7)(441 - 196)$ (1) $= (22/7) \times 245 = 770\ m^2$ (2)

Cost = 770 × 50 = ₹ 38500 (1)

SECTION E — Case-Based / Integrated Questions 3 × 4 = 12

36. Case Study 1 — Taxi fare

A taxi service charges a fixed amount of ₹ 50 plus ₹ 12 for every kilometre travelled. Let x be the distance travelled in km.

(i) Write the fare as a linear polynomial in x.

Fare = $12x + 50$ (1)

(ii) Find the fare for a journey of 8 km.

$12(8) + 50 = 96 + 50 =$ ₹ 146 (1)

(iii) Ravi paid ₹ 230. How many kilometres did he travel?

$12x + 50 = 230$ (½)

$12x = 180$ (1)

$x = 15$ km (½)

37. Case Study 2 — Favourite sport

In a survey of 50 students, 20 chose cricket, 15 chose football, 10 chose hockey and 5 chose kabaddi as their favourite sport. One student is selected at random.

(i) Find the probability that the student chose cricket.

$20/50 = 2/5$ (1)

(ii) Find the probability that the student did not choose football.

$(50 - 15)/50 = 35/50 = 7/10$ (1)

(iii) Find the probability that the student chose hockey or kabaddi.

Favourable = 10 + 5 = 15 (1)

P = 15/50 = 3/10 (1)

38. Case Study 3 — Monthly savings

Riya saves ₹ 200 in the first month and increases her saving by ₹ 50 every month.

(i) Write the first four terms of the sequence of her monthly savings.

200, 250, 300, 350 (1)

(ii) How much does she save in the 12th month?

$t_{12} = 200 + 11 \times 50 = ₹\,750$ (1)

(iii) Find her total savings in the first 12 months.

$S_{12} = (12/2)[2(200) + 11(50)]$ (1)

$= 6 \times 950 = ₹\,5700$ (1)

— End of Answer Key —